

Kynurenine Pathway Dynamics and Affective Response to High- and Low-Intensity Resistance Exercise: A Randomized Crossover Trial

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ABSTRACT 253

01 INTRODUCTION

The **kynurenine pathway (KP)** produces metabolites involved in inflammation, immune response and excitatory neurotransmission¹. KP is activated by acute physical exercise, metabolizing **kynurenine (KYN)** to either **kynurenic acid (KA)** or **quinolinic acid (QA)**². The ratios KA/KYN (neuroprotective index), QA/KYN (neurotoxic index) and QA/KA, thus, reflect the relative activity of these branches.

Aim: KP dynamics are well described for endurance exercise, evidence for **resistance exercise (RE)** is limited. It remains unclear whether RE intensity modulates KP responses and whether these responses relate to affect.

02 METHODS

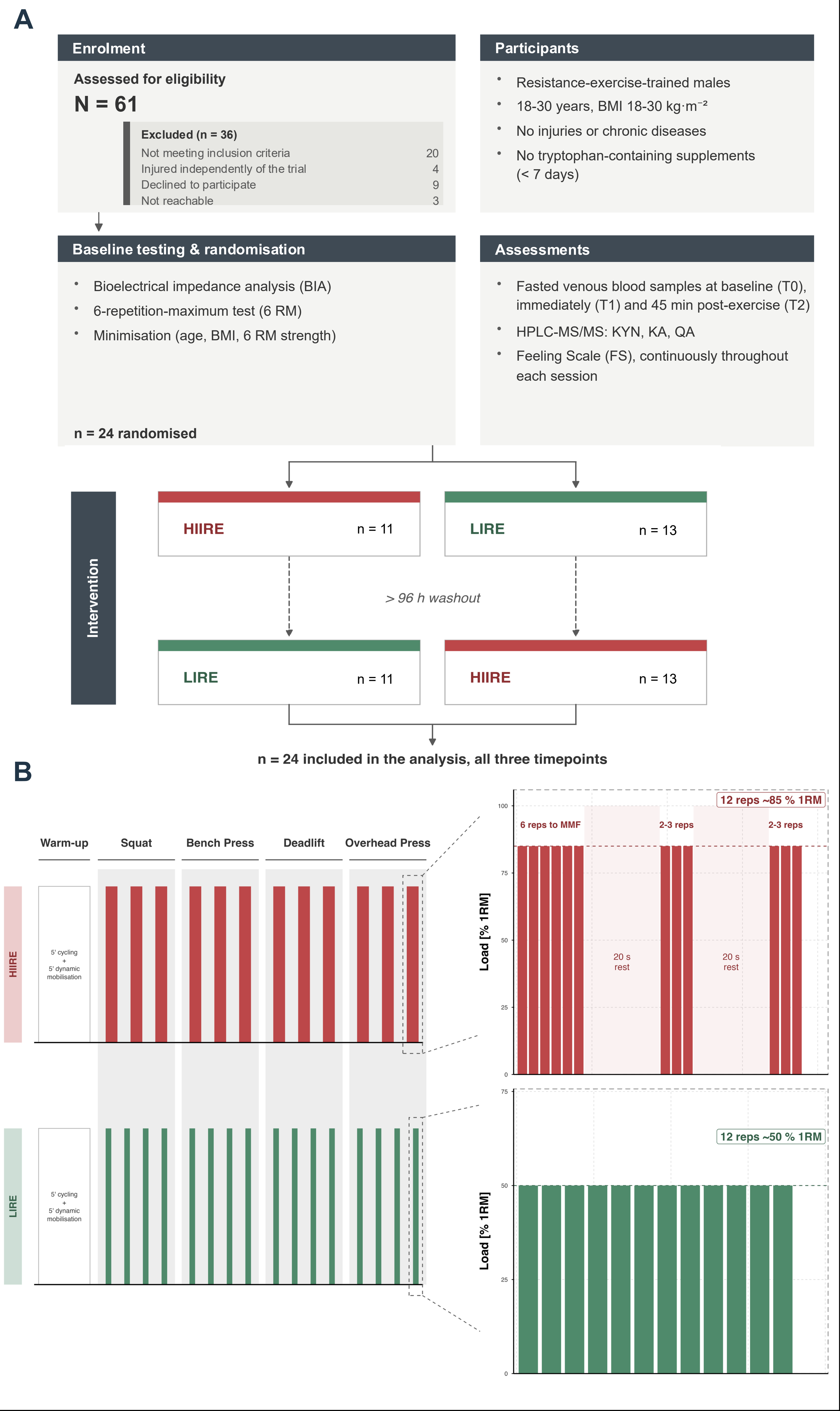


Fig. 1. Study design and RE protocol.
A. BMI body mass index, HPLC-MS/MS High-performance liquid chromatography–tandem mass spectrometry.
B. Both protocols were matched for total relative volume and session duration. MMF Momentary muscle failure, RM repetition maximum

03 RESULTS

Tab. 1. Participant characteristics³

	Total	HIIRE+LIRE	LIRE+HIIRE	p ^a
	n = 24	n = 11	n = 13	
Age	23.50 ± 2.90	23.00 ± 2.10	23.80 ± 3.40	.491
Body mass (kg)	83.85 ± 10.09	86.62 ± 9.87	81.52 ± 10.05	.225
Height (m)	1.82 ± 0.07	1.82 ± 0.08	1.81 ± 0.07	.694
Fat mass (kg)	15.22 ± 6.00	16.77 ± 4.59	13.91 ± 6.87	.252
Fat mass (%)	17.82 ± 5.78	19.19 ± 3.75	16.67 ± 7.01	.297
FFM [kg]	68.19 ± 7.13	69.06 ± 7.15	67.45 ± 7.32	.595
Total 6RM Score (min. 4 max. 12)	8.80 ± 1.60	8.60 ± 1.40	8.90 ± 1.70	.664

HIIRE: High-Intensity Interval Resistance Exercise, LIRE: Low-Intensity Resistance Exercise, FFM: Fat-Free mass, RM: Repetition Maximum, Means ± SDs are presented, ^aIndependent Samples T-Test

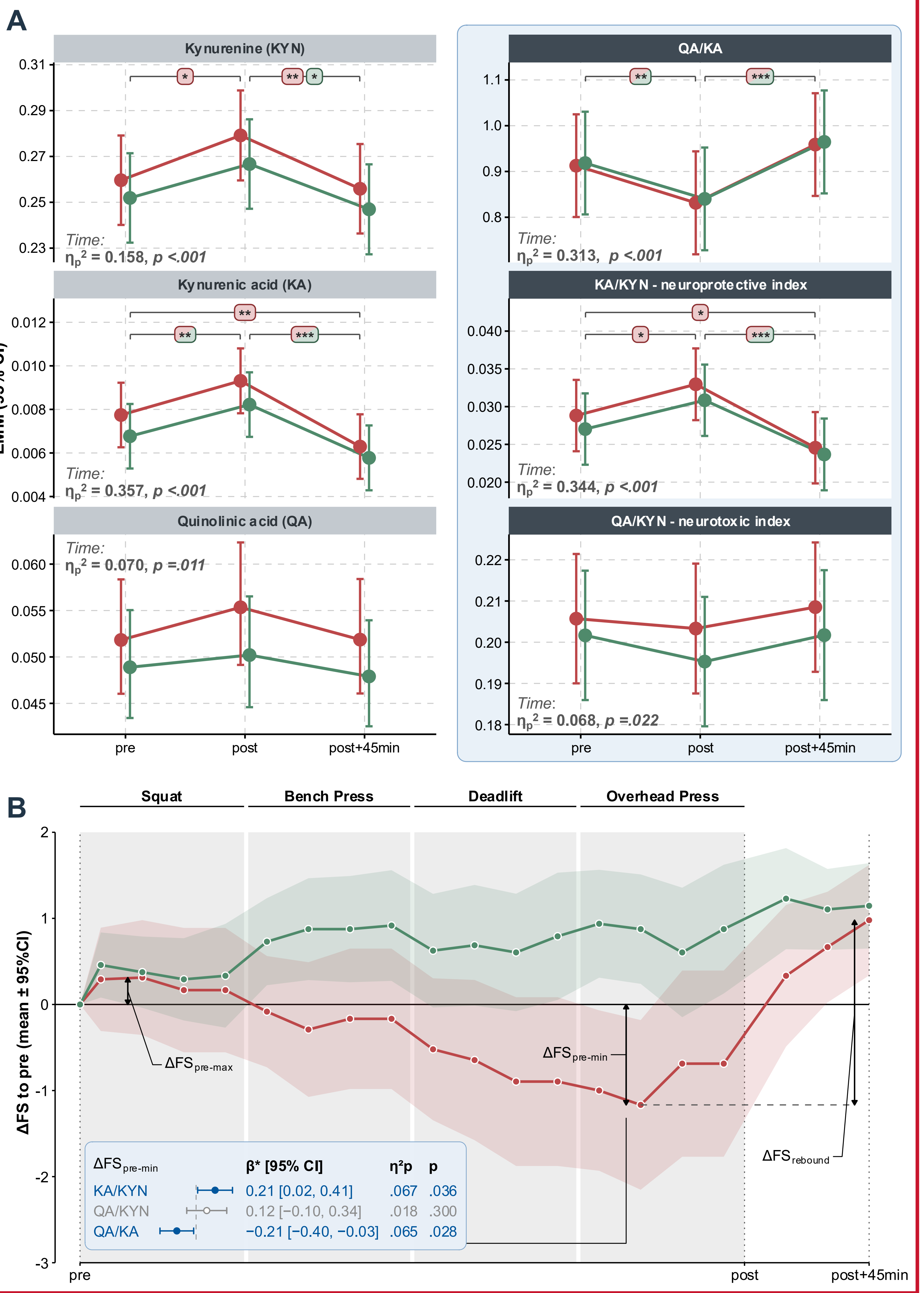


Fig. 2. Kinetics of kynurenine pathway metabolite ratios and time course of Feeling Scale

04 CONCLUSION

- 01.** ΔFS_{pre-min} emerged as a marker of acute KP activation. Greater subjective stress load was related to a stronger neuroprotective shift.
- 02.** Despite marked protocol differences, **KP responses were similar**, indicating not differentially modulated by RE intensity alone.



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¹Cervenka, I., Agudelo, L. Z., & Ruas, J. L. (2017). Kynurenines: Tryptophan's metabolites in exercise, inflammation, and mental health. *Science*, 357(6349). <https://doi.org/10.1126/science.aaf9794>
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